Similarity & Proportions Maze

Name _____ Period ____

Start in the top-left room. Solve, go through the door with your answer, and show your work on the answer grid. Shadows are measured at the same time.

START

Solve.

$\frac{x}{4} = \frac{9}{12}$

$x = 3$

Solve.

$\frac{5}{x} = \frac{15}{6}$

$x = \frac{25}{2}$

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$. Find EC.

$AD = 2$, $DB = 7$

$AE = 4$

$EC = \frac{8}{7}$

$\triangle JKL \sim \triangle MNO$. Find NO.

$JK = 12$, $MN = 8$

$KL = 15$

$x = \frac{16}{3}$

$x = 27$

$x = 2$

$x = 18$

$x = \frac{28}{3}$

$EC = 15$

$EC = 8$

Find the tower's height.

5-ft pole: 3-ft shadow

tower: 27-ft shadow

$k = 3$

A dilation maps $\triangle ABC$ onto $\triangle DEF$. Find the scale factor k.

$AB = 6$, $DE = 9$

$k = \frac{2}{3}$

Solve.

$\frac{x + 2}{5} = \frac{6}{3}$

$x = 8$

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$. Find EC.

$AD = 4$, $DB = 6$

$AE = 6$

$\frac{81}{5}$ ft

29 ft

$k = \frac{3}{2}$

$EF = 15$

$x = 10$

$EC = 4$

$EC = 9$

Solve.

$\frac{7}{3} = \frac{x}{9}$

$x = \frac{27}{7}$

$\triangle ABC \sim \triangle DEF$. Find EF.

$AB = 4$, $DE = 10$

$BC = 6$

$EF = 12$

$\triangle PQR \sim \triangle XYZ$. Find QR.

$PQ = 10$, $XY = 4$

$YZ = 8$

33 ft

Find the tree's height.

6-ft man: 4-ft shadow

tree: 22-ft shadow

$x = 63$

$EF = 24$

$EF = \frac{12}{5}$

$QR = 20$

$QR = \frac{16}{5}$

24 ft

$\frac{44}{3}$ ft

A dilation maps $\triangle RST$ onto $\triangle UVW$. Find the scale factor k.

$RS = 20$, $UV = 8$

$x = 1$

Solve.

$\frac{x}{x + 3} = \frac{4}{7}$

$x = 4$

$\triangle ABC \sim \triangle DEF$. Find x.

$AB = 8$, $BC = 10$

$DE = 12$, $EF = 2x + 3$

$x = 6$

FINISH

You made it!

Similarity & Proportions Maze: Answer Grid

Name _____ Period ____

Write each answer in the order you found it, and show your work for that room.

1	Answer: _____	Work:
2	Answer: _____	Work:
3	Answer: _____	Work:
4	Answer: _____	Work:
5	Answer: _____	Work:
6	Answer: _____	Work:
7	Answer: _____	Work:
8	Answer: _____	Work:
9	Answer: _____	Work:
10	Answer: _____	Work:

Similarity & Proportions Maze ANSWER KEY

Start in the top-left room. Solve, go through the door with your answer, and show your work on the answer grid. Shadows are measured at the same time.

START

1 Solve.

$\frac{x}{4} = \frac{9}{12}$

$x = 3$

2 Solve.

$\frac{5}{x} = \frac{15}{6}$

$x = \frac{25}{2}$

In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$. Find EC.

$AD = 2$, $DB = 7$

$AE = 4$

$EC = \frac{8}{7}$

$\triangle JKL \sim \triangle MNO$. Find NO.

$JK = 12$, $MN = 8$

$KL = 15$

$x = \frac{16}{3}$

$x = 27$

$x = 2$

$x = 18$

$x = \frac{28}{3}$

$EC = 15$

$EC = 8$

Find the tower's height.

5-ft pole: 3-ft shadow

tower: 27-ft shadow

$\frac{81}{5}$ ft

29 ft

3 A dilation maps $\triangle ABC$ onto $\triangle DEF$. Find the scale factor k.

$AB = 6$, $DE = 9$

$k = 3$

$k = \frac{3}{2}$

5 Solve.

$\frac{x+2}{5} = \frac{6}{3}$

$x = 8$

$x = 10$

6 In $\triangle ABC$, $\overline{DE} \parallel \overline{BC}$. Find EC.

$AD = 4$, $DB = 6$

$AE = 6$

$EC = 4$

$EC = 9$

Solve.

$\frac{7}{3} = \frac{x}{9}$

$x = 63$

$x = \frac{27}{7}$

EF = 24

4 $\triangle ABC \sim \triangle DEF$. Find EF.

$AB = 4$, $DE = 10$

$BC = 6$

$EF = \frac{12}{5}$

EF = 15

$EF = 12$

8 $\triangle PQR \sim \triangle XYZ$. Find QR.

$PQ = 10$, $XY = 4$

$YZ = 8$

$QR = \frac{16}{5}$

QR = 20

33 ft

24 ft

7 Find the tree's height.

6-ft man: 4-ft shadow

tree: 22-ft shadow

$\frac{44}{3}$ ft

A dilation maps $\triangle RST$ onto $\triangle UVW$. Find the scale factor k.

$RS = 20$, $UV = 8$

$x = 1$

9 Solve.

$\frac{x}{x+3} = \frac{4}{7}$

$x = 4$

10 $\triangle ABC \sim \triangle DEF$. Find x.

$AB = 8$, $BC = 10$

$DE = 12$, $EF = 2x + 3$

$x = 6$

FINISH

You made it!

Similarity & Proportions Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **$x = 3$**	$12x = 36 \rightarrow x = 3$
2	Answer: **$x = 2$**	$15x = 30 \rightarrow x = 2$
3	Answer: **$k = \frac{3}{2}$**	$k = \frac{DE}{AB} = \frac{9}{6} \rightarrow k = \frac{3}{2}$
4	Answer: **$EF = 15$**	$\frac{EF}{6} = \frac{10}{4} \rightarrow 4 \cdot EF = 60 \rightarrow EF = 15$
5	Answer: **$x = 8$**	$3(x + 2) = 30 \rightarrow x + 2 = 10 \rightarrow x = 8$
6	Answer: **$EC = 9$**	$\frac{4}{6} = \frac{6}{EC} \rightarrow 4 \cdot EC = 36 \rightarrow EC = 9$
7	Answer: **33 ft**	$\frac{h}{22} = \frac{6}{4} \rightarrow 4h = 132 \rightarrow h = 33$
8	Answer: **$QR = 20$**	$\frac{QR}{8} = \frac{10}{4} \rightarrow 4 \cdot QR = 80 \rightarrow QR = 20$
9	Answer: **$x = 4$**	$7x = 4(x + 3) \rightarrow 7x = 4x + 12 \rightarrow 3x = 12 \rightarrow x = 4$
10	Answer: **$x = 6$**	$\frac{2x + 3}{10} = \frac{12}{8} \rightarrow 2x + 3 = 15 \rightarrow x = 6$